

Organize the Freezer SOP (1x Freezer)

One episode organizes one countertop freezer. It runs from a freezer holding six items lying loose in a jumble in front of two empty bins, to a freezer holding the two bins seated square at the back wall, each bin holding its own category stacked with the oldest item on top, two bags carrying a fresh label, and the door pressed fully shut.

The order never changes: **pull all six items out, relabel the two named bags, draw both bins forward, load each bin oldest up, slide both bins home, then close the door.** The items come out before the bins move, so nothing is carried over a bin. The relabeling happens while the bags still lie flat on the table, because a bag buried in a stack cannot be labeled.

Every bin is loaded newest first and oldest last, so the oldest item ends on top. In each bin the item marked **3** goes in first, then **2**, then **1** on top. This is what "oldest up" means everywhere in this document, and it is the same in both bins.

Arm roles are fixed and never swap. The **right gripper is the item gripper**: it pulls every item out of the freezer, lays each one on its sort spot, peels and presses both labels, lays every item into its bin, and presses the door fully shut at the end. The **left gripper is the bin and door gripper**: it draws both bins forward, slides them home, holds the label sheet flat, holds a bag flat during a label press, and swings the door round at the end. Neither arm reaches across the body.

The door is hinged on the left of the freezer. It stands open against the **door post** for the whole of the work, so the freezer mouth stays clear for the right gripper.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the freezer and its frame includes the freezer mouth, the whole freezer floor, both bins seated and drawn forward, the door open on the post, the label sheet, the bag spot, the box spot, and the front table edge.
3. The camera reads the freezer floor from the front, so which items are still inside, where each bin sits, and whether anything sticks out past the freezer front lip are all readable.

4. The camera reads each item's date number when the item lies on its sort spot, so which item is which is readable.
5. The camera reads each loaded bin from above, so the order of the stack and the date number on the top item are readable.
6. The camera reads the shut door from the front, so daylight along the seal is readable.
7. Both arms are at home with grippers open.
8. The freezer is steady on the table and does not slide when a bin is drawn forward or the door is pressed shut.
9. The right arm reaches every item on the freezer floor, both bins when they stand at the freezer front lip, the label sheet, the bag spot, the box spot, and the door's front face, without extending to a joint limit. It reaches neither the door post nor the door's free edge.
10. The left arm reaches both bins at the back wall and at the front lip, the label sheet, the bag spot, and the door's free edge through its whole swing, without extending to a joint limit. It reaches neither the box spot nor the freezer floor items.
11. Each arm comes onto the freezer floor straight in from the front and comes down onto a bin straight from above, without its wrist or forearm touching the freezer roof, the freezer mouth, or the door.
12. The two arms do not collide at the freezer, and the open door is clear of the right arm's path to the freezer mouth.
13. If a place cannot be reached, move it toward the arm that cannot reach it until lines 9 to 12 hold.

Materials checklist

1. The **freezer** is a countertop freezer standing on the table at the back center, squared to the back table edge, its mouth facing the front. It is heavy enough that it does not move when a bin is drawn forward or the door is pressed shut.
2. The freezer is one compartment with one flat **freezer floor**. There is no shelf and no drawer, and there is nothing else inside it.
3. The freezer floor is deep enough front to back for a bin to stand at the back wall and to stand again at the freezer front lip, and wide enough for the two bins to stand side by side with clear space between them.
4. There is clear space above both bins, so a loaded bin stands on the floor without its stack touching the freezer roof.
5. **Two bins** stand on the freezer floor, side by side, seated square against the back wall, with clear space between them. Both start empty.
6. The **bag bin** is the left one and carries a blue **BAGS** label on its front rail. The **box bin** is the right one and carries a green **BOXES** label on its front rail.

7. Each bin is open topped, rigid, and light. Each has a **front rail** — a raised top edge along its front wall that a gripper closes on. A bin slides on the freezer floor when drawn by its front rail and does not need to be lifted.
8. Each bin is deep enough to hold three items stacked flat with the top item sitting below the bin rim, and is no wider than one item, so a stack has one place to go and no choice of lane.
9. **Six items** lie flat on the freezer floor in front of the bins, in one row across the floor, none touching another, none on top of another, each with its date number facing up. They lie in this fixed order, left to right: **box 2, bag 3, box 1, bag 1, box 3, bag 2**.
10. **Three items are bags** and **three items are boxes**. A bag is a flat sealed freezer bag that keeps its shape when gripped and lies flat on one face. A box is a rigid carton that lies flat on one face.
11. Every item carries a **date tag** on the face that starts upward, printed with one **date number**: 1, 2, or 3. Each category uses each number once, so there is a bag 1, a bag 2, a bag 3, a box 1, a box 2, and a box 3.
12. Every item is closed, dry, and sealed, and is held by one gripper without its contents spilling or its seal opening.
13. Every bag carries a blank **label patch** on the same face as its date tag, beside the date tag and not overlapping it.
14. One **label sheet** lies flat on the table at the front, to the left of the bag spot, written side up, with no corner lifted.
15. The label sheet carries **two labels** side by side. The **left label names bag 2** and the **right label names bag 3**. Each label is written large enough for the camera to read the bag it names.
16. The **sort strip** is the clear table across the front. It holds the **bag spot** at the front center and the **box spot** at the front right, side by side and not touching, and nothing else.
17. Each sort spot is large enough for three items to lie flat side by side with clear space between them.
18. The **door** is hinged on the freezer's left front edge. It swings open to the left and it is standing open against the **door post** at the start of the episode.
19. The **door post** is a heavy fixed block at the back left of the table. The open door rests its outer face against it and stands steady there.
20. The door shuts with a latch that holds it closed, and its seal runs all round the mouth. A correctly shut door shows no daylight along the seal and does not spring back open.
21. The table is clean, dry, and bare apart from the freezer, the door post, and the label sheet.

Workspace layout

- **Freezer:** the countertop freezer at the back center. It is never lifted, never turned, and never pushed out of its place.

- **Freezer mouth:** the open front of the freezer. Every gripper going inside passes through it straight in from the front.
- **Freezer floor:** the one flat floor inside the freezer. The bins stand on it and the six items start on it.
- **Freezer front lip:** the front edge of the freezer floor. Nothing may stick out past it, or the door will not shut.
- **Bag bin:** the left bin, labeled **BAGS** in blue. It takes the three bags. **Left gripper moves it; right gripper loads it.**
- **Box bin:** the right bin, labeled **BOXES** in green. It takes the three boxes. **Left gripper moves it; right gripper loads it.**
- **Sort strip:** the clear table across the front, holding the two sort spots.
- **Bag spot:** the front center of the sort strip. Every bag lands here after it is pulled. **Right gripper lays items here; left gripper only holds a bag flat during a label press.**
- **Box spot:** the front right of the sort strip. Every box lands here after it is pulled. **Right gripper only.**
- **Label sheet:** flat on the table at the front, to the left of the bag spot, written side up. It stays flat on the table for the whole episode.
- **Door:** the freezer door, hinged on the left, standing open against the door post for all the work.
- **Door post:** the heavy fixed block at the back left of the table that the open door rests against. It is never moved.

Arm assignments

- **Right gripper — the item gripper.** Pulls each of the six items off the freezer floor, lays each one on its sort spot, peels each label off the label sheet and presses it onto its bag, lays each item into its bin, and presses the door fully shut at the end. It never touches a bin, the door post, or the door's free edge.
- **Left gripper — the bin and door gripper.** Draws each bin forward to the freezer front lip, slides each bin home to the back wall, holds the label sheet flat during a peel, holds a bag flat during a label press, and swings the door off the post and round to the freezer at the end. It never carries an item and never lays an item into a bin.
- Nothing is handed between grippers, and only one arm moves an object at a time.

Vocabulary

- **Date number:** the number printed on an item's date tag — 1, 2, or 3. **1 is the oldest and 3 is the newest.** There is one of each number in each category.
- **Oldest up:** the item marked 1 ends on top of that bin's stack. It is reached by laying 3 in first, then 2 on top of it, then 1 on top of that. The same order is used in both bins.

- **Stack:** the items lying flat one on top of another in a bin, all facing up, all lined up with the bin walls.
- **Number up:** the item lies on the face opposite its date tag, so the date number faces the ceiling and the camera can read it.
- **Lies flat:** the item rests on one whole face, not on an edge, not leaning, and not propped on another item.
- **Front rail:** the raised top edge along a bin's front wall. It is the only part of a bin a gripper closes on.
- **Seated:** the bin stands on the freezer floor with its back against the freezer back wall and its walls lined up with the freezer walls, not turned and not tilted.
- **Square:** lined up with the walls it sits against, not turned to one side.
- **Drawn forward:** the bin has been slid straight toward the front along the freezer floor until its front rail stands just inside the freezer front lip, still lined up square.
- **Stands proud:** part of an item sits above the bin rim, or part of a bin or an item sticks out past the freezer front lip.
- **Clear space:** two things do not touch each other, and the gap between them is visible from the front.
- **Label patch:** the blank rectangle printed on a bag's face beside its date tag. A new label goes here and nowhere else.
- **Label sheet:** the backing sheet carrying the two labels, lying flat on the table at the front, written side up.
- **Label face:** the written side of a label, naming the bag it belongs to by that bag's date number.
- **Steady hold:** the left gripper closed and pressed down on a bag lying flat on the table, held still for the whole of the label press so the bag does not slide.
- **Sealed:** the door face meets the freezer all round, no daylight shows anywhere along the seal, the latch holds, and the door does not spring back when the gripper comes off.

Steps

Step 1: Pull the six items and sort them onto the strip

Goal: the freezer floor holds nothing but the two seated bins, and all six items lie flat on the sort strip, bags on the bag spot and boxes on the box spot, every date number facing up.

Work the items from the left end of the row to the right end, one at a time. Do not skip one and come back to it.

1.1 Take one item off the freezer floor

- With the **right gripper**, come in straight from the front through the freezer mouth, level, above the next item in the row.
- Close on the **body** of the item, across its middle, not on a corner and not on its seal.
- Lift the item straight up clear of the other items, then carry it straight out to the front through the freezer mouth, level and low.
- Do not pass over a bin, and do not let the item touch another item on the way out.

1.2 Lay the item on its sort spot

- Read the item as it comes out: a **bag** goes to the **bag spot**, a **box** goes to the **box spot**.
- With the **right gripper**, lower the item onto its spot until it rests flat on the table, then open the gripper.
- Lay it **number up**, beside the items already on that spot, with clear space between them, none touching and none on top of another.
- Repeat 1.1 and 1.2 until all six items are out.

Check: the freezer floor holds nothing but the two bins, still seated square at the back wall. Three bags lie flat on the bag spot and three boxes lie flat on the box spot, every date number facing up and readable, none touching, none stacked. If an item is on the wrong spot, move it with the **right gripper** to the correct spot. If an item lies number down, turn it over with the **right gripper**. If an item is still inside, pull it now.

Expected state: freezer floor clear apart from the two seated bins, six items laid out on the sort strip in two groups of three.

Step 2: Relabel the two named bags

Goal: bag 2 and bag 3 each carry a fresh label pressed flat in their label patch.

Do the **left label first**, then the right one. The bags are labeled now, while they still lie flat on the table, because a bag in a stack cannot be labeled.

2.1 Peel one label

- With the **left gripper**, close on the near corner of the **label sheet** and hold the sheet flat on the table.
- With the **right gripper**, close on the corner of the **left label** and peel it off the sheet in one motion, label face out.
- Read the **label face** and find the bag on the bag spot whose date number matches it.
- The sheet stays flat on the table for the whole of the peel. If a label tears or folds onto itself, end the episode (see Failures that are not violations).

2.2 Press the label onto its bag

- The **left gripper** releases the sheet and takes a **steady hold** on the named bag, holding it flat on the bag spot, while the **right gripper** holds the label clear of the table.
- With the **right gripper**, bring the label down onto that bag's **label patch**, label face up and the writing upright.
- Press it down, then press along it from one edge to the other until it lies flat with no lifted corner.
- Do not let the label cover the date tag.
- Open both grippers and draw both arms clear.

2.3 Repeat for the second label

- Repeat 2.1 and 2.2 with the **right label** on the sheet and the bag it names.

Check: two bags carry a new label, each label sits in that bag's label patch, each label names that bag's own date number, the writing is upright, each label lies flat with no lifted corner, no label covers a date tag, every date number is still readable, and the label sheet still lies flat on the table with no labels left on it. If a label lifts at a corner, press it flat again with the **right gripper**. If a label is on the wrong bag, end the episode (see Failures that are not violations).

Expected state: six items still lying flat on the sort strip, two of the bags newly labeled, label sheet empty and flat.

Step 3: Draw both bins forward

Goal: both bins stand at the freezer front lip, square, ready to be loaded from above.

The bins are drawn forward so the right gripper can lay items straight down into them without reaching deep into the freezer and without passing over the other bin.

- With the **left gripper**, close on the **front rail** of the **bag bin**.
- Draw the bin straight toward the front along the freezer floor until its front rail stands just inside the **freezer front lip**. Keep it square as it comes.
- Do not lift the bin, do not turn it, and do not let it stand proud of the freezer front lip.
- Open the **left gripper** and draw it back clear.
- Repeat with the **front rail** of the **box bin**.

Check: both bins stand at the freezer front lip, square, side by side with clear space between them, neither one standing proud of the lip, and both still empty. If a bin sits turned, square it up with the **left gripper**. If a bin stands proud of the lip, push it back with the **left gripper** until it sits just inside.

Expected state: two empty bins drawn forward and square at the freezer front lip.

Step 4: Load each bin oldest up

Goal: the bag bin holds the three bags and the box bin holds the three boxes, each stacked flat and square with the item marked 1 on top.

Load the **bag bin first, then the box bin**, so nothing is ever carried over a loaded bin. Within each bin the order is always **3, then 2, then 1**.

4.1 Load the bag bin

- With the **right gripper**, close on the **body** of **bag 3** on the bag spot, across its middle, not on a corner and not on its seal.
- Lift it, carry it level to the **bag bin**, and bring it down straight from above.
- Lower it until it rests flat on the bin floor, lined up with the bin walls and **number up**, then open the gripper.
- Repeat with **bag 2**, laying it flat on top of bag 3, lined up with it and number up.
- Repeat with **bag 1**, laying it flat on top of bag 2, lined up with it and number up.

4.2 Load the box bin

- Repeat 4.1 with the boxes into the **box bin**: **box 3** first, then **box 2** on top of it, then **box 1** on top of that, each laid flat, lined up with the bin walls, and number up.
- Do not carry a box over the loaded bag bin. Come to the box bin from the front and down from above.

Check: each bin holds three items of one category only, stacked flat and lined up with the bin walls. The top item of each bin shows the number **1**. No item stands proud of its bin rim, none sits on edge or leans, and both sort spots are empty. If a stack is out of line, lift the items above it with the **right gripper** and lay them down again. If an item is in the wrong bin, move it with the **right gripper** to the correct bin and rebuild both stacks in the order 3, 2, 1.

Expected state: bag bin holding three bags oldest up, box bin holding three boxes oldest up, sort strip empty.

Step 5: Reseat both bins

Goal: both bins stand seated square against the freezer back wall, with nothing standing proud of the freezer front lip.

- With the **left gripper**, close on the **front rail** of the **bag bin**.
- Push the bin straight back along the freezer floor until it meets the freezer back wall. Keep it square as it goes.
- Do not lift the bin, do not turn it, and do not push after it has met the back wall.
- Open the **left gripper** and draw it back clear.
- Repeat with the **front rail** of the **box bin**.

Check: both bins are seated square against the back wall, side by side with clear space between them. Nothing sticks out past the freezer front lip, and no item stands proud of a bin rim. Each stack is still in its order with the number 1 on top. If a bin sits turned or short of the back wall, push it home with the **left gripper**. If an item has shifted in a stack, lay it flat again with the **right gripper**.

Expected state: two loaded bins seated square at the back wall, freezer floor otherwise clear.

Step 6: Close the door fully

Goal: the door is shut and sealed, with no daylight along the seal and no spring back.

6.1 Swing the door round

- Check first that both arms are clear of the freezer mouth and that nothing sticks out past the freezer front lip.
- With the **left gripper**, close on the door's free edge at about its middle height.
- Swing the door away from the door post and round toward the freezer in one smooth move. Do not swing it fast, and do not let it hit the freezer.
- Stop when the door has come round in front of the freezer mouth and its free edge is near the middle of the freezer front.
- Open the **left gripper** and draw it back clear of the door.

Check: the door hangs in front of the freezer mouth, nothing is caught between the door and the freezer, and no bin has been knocked out of place. If a bin was knocked, swing the door open again with the **left gripper** and push the bin home with the **left gripper** before going on.

6.2 Press the door home

- With the **right gripper** closed, press the door's front face flat, at the free edge side, straight back toward the freezer.
- Press until the door meets the freezer all round and the latch holds. Hold the press for 2 seconds.
- Press straight back only. Do not slam the door, and do not press hard enough to move the freezer.
- Draw the **right gripper** straight back off the door.

Check: the door is sealed — it meets the freezer all round, no daylight shows anywhere along the seal, and the door does not spring back when the **right gripper** comes off. If daylight shows or the door springs back, press it home once more with the **right gripper**. If it still will not seal, swing it open with the **left gripper** and look for a bin or an item standing proud of the freezer front lip.

Expected state: door shut and sealed, freezer square to the back table edge, sort strip empty, label sheet empty and flat, door post still in its place.

Step 7: End the episode

- Confirm the door: shut and sealed, no daylight along the seal, and no spring back.
- Confirm the table: the bag spot is empty, the box spot is empty, the label sheet lies flat with no labels left on it, and nothing is left on the table but the freezer, the door post, and the label sheet.
- Confirm the freezer: still square to the back table edge, and the door post still in its place.
- Confirm nothing is on the floor or lying on the table beside the freezer.
- Correct any failed check before ending.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Swing the door open by hand and rest it against the door post, outer face on the post.
2. Draw both bins forward by hand and take the three bags out of the bag bin and the three boxes out of the box bin.
3. Push both bins back until they are seated square against the freezer back wall, side by side with clear space between them, and both empty.
4. Lay the six items back on the freezer floor in front of the bins, in one row, flat, number up, none touching, in this order left to right: box 2, bag 3, box 1, bag 1, box 3, bag 2.
5. Take the two labels off the two labeled bags and clear their label patches, so every label patch is blank again.
6. Lay a fresh label sheet flat on the table to the left of the bag spot, written side up with no corner lifted, carrying two labels: the left one naming bag 2, the right one naming bag 3.
7. Clear the bag spot and the box spot, and wipe and dry any spill on the table, on the freezer floor, or in a bin.
8. Replace an item that leaks, whose seal has opened, that will not lie flat on one face, or whose date tag can no longer be read.
9. Replace a bag whose label patch will no longer hold a label.
10. Replace a bin whose front rail is damaged, that no longer slides on the freezer floor, or that no longer stands square against the back wall.
11. Check the door still latches and seals when it is pushed shut by hand, and that it does not spring back.
12. Check the freezer still stands square to the back table edge and does not slide when a bin is drawn forward.
13. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong work order

- **Visible cue:** a bin is drawn forward before all six items are out, a bag is labeled after it has gone into a bin, a bin is loaded before both bins are drawn forward, the box bin is loaded before the bag bin is finished, a bin is reseated before both bins are loaded, or a phase is left part done and come back to.
- **SOP rule broken:** Steps 1 to 5, pull all six items, relabel the two bags, draw both bins forward, load the bag bin then the box bin, then reseat both bins.
- **Coaching note:** items out, labels on, bins forward, bins loaded, bins home. Finish each phase before the next one starts.

Violation: Items left in the freezer

- **Visible cue:** the episode moves on to Step 2 with an item still lying on the freezer floor, or a bin is drawn forward over an item still on the floor.
- **SOP rule broken:** Step 1, all six items come out of the freezer before anything else happens.
- **Coaching note:** count the sort spots before you move on. Three on each is the whole six.

Violation: Item sorted to the wrong spot

- **Visible cue:** the **right gripper** lays a bag on the box spot or a box on the bag spot, or an item ends the step on the spot for the other category.
- **SOP rule broken:** Step 1.2, a bag goes to the bag spot and a box goes to the box spot.
- **Coaching note:** read the item as it comes out, then choose the spot. Sorting on the way out is what makes the loading easy.

Violation: Item not laid flat or number not up

- **Visible cue:** an item on a sort spot rests on an edge, leans, sits propped on another item, sits on top of another item, touches a neighbour, or lies with its date tag against the table so the number cannot be read.
- **SOP rule broken:** Step 1.2, lay every item flat on its spot, number up, with clear space between items and none stacked.
- **Coaching note:** flat, apart, number up. If the camera cannot read the number, the sort cannot be checked.

Violation: Item gripped by a corner or a seal

- **Visible cue:** the **right gripper** closes on the corner of a bag or box, on a bag's seal, or on a bag's label patch instead of across the item's body, or a seal opens as the item is carried.
- **SOP rule broken:** Steps 1.1 and 4.1, close on the body of the item, across its middle, not on a corner and not on its seal.
- **Coaching note:** grip across the middle where the item is solid. A corner grip is how a bag comes open.

Violation: Label sheet not held down

- **Visible cue:** the **left gripper** is off the label sheet during a peel, or the sheet lifts, slides, or turns as the **right gripper** pulls the label.
- **SOP rule broken:** Step 2.1, the **left gripper** holds the sheet flat on the table for the whole of the peel.
- **Coaching note:** hold the sheet down first, then peel. A sheet that lifts tears the label.

Violation: Label put on the wrong bag or the wrong place

- **Visible cue:** a label ends up on a bag whose date number does not match the label face, on a box, outside the label patch, overlapping the date tag, or with the writing upside down.
- **SOP rule broken:** Steps 2.1 and 2.2, read the label face, find the bag whose date number matches, and press the label into that bag's label patch with the writing upright and the date tag left clear.
- **Coaching note:** read the label, then find the bag. Never pick the bag first.

Violation: Label press done wrong

- **Visible cue:** the **left gripper** is off the bag while the **right gripper** presses the label, or the bag slides, lifts, or turns during the press; or the label ends the step with a lifted corner, a bubble, or a fold because the **right gripper** put it down and moved away without pressing along it.
- **SOP rule broken:** Step 2.2, the **left gripper** takes a **steady hold** on the bag and holds it flat for the whole of the press, and the **right gripper** presses the label down and then along it from one edge to the other until it lies flat with no lifted corner.

- **Coaching note:** hold the bag flat before the label comes down, then press it down and press it along. A lifted corner peels off in the cold.

Violation: Bin lifted or turned instead of slid

- **Visible cue:** the **left gripper** lifts a bin off the freezer floor, carries it in the air, turns it, or drags the freezer with it.
- **SOP rule broken:** Steps 3 and 5, close on the bin's front rail and slide it straight along the freezer floor, keeping it square, without lifting it.
- **Coaching note:** rail, straight, square. A bin never leaves the floor.

Violation: Bin not drawn to the front lip or left standing proud

- **Visible cue:** a bin is loaded while it still stands at the back wall or only part way forward, or a bin ends Step 3 sitting turned or with part of it past the freezer front lip.
- **SOP rule broken:** Step 3, draw each bin forward until its front rail stands just inside the freezer front lip, square and not proud of the lip.
- **Coaching note:** bring the bin to you before you load it, and stop it just inside the lip.

Violation: Item put in the wrong bin

- **Visible cue:** a bag is laid into the box bin or a box into the bag bin, or a bin ends Step 4 holding items of both categories.
- **SOP rule broken:** Step 4, the three bags go into the bag bin and the three boxes go into the box bin.
- **Coaching note:** read the bin rail label and the item before the arm comes down. BAGS takes bags only.

Violation: Oldest not ended on top

- **Visible cue:** a bin is loaded in an order other than 3, then 2, then 1, or the top item of a finished bin shows a number other than 1.
- **SOP rule broken:** Step 4, load each bin newest first and oldest last — 3, then 2, then 1 on top.
- **Coaching note:** count down as you load: three, two, one. The one you see from above is always the oldest.

Violation: Stack not square or standing proud of the bin rim

- **Visible cue:** items in a bin are not lined up with the bin walls or with each other, an item sits on edge or leans in the bin, an item lies number down, or the top item sits above the bin rim.
- **SOP rule broken:** Step 4, lay each item flat, lined up with the bin walls, number up, with no item standing proud of the bin rim.

- **Coaching note:** line each one up with the walls as it goes down. A crooked stack is a lid that will not clear.

Violation: Item dropped into a bin

- **Visible cue:** the **right gripper** opens above a bin and the item falls in, the item bounces or shifts on landing, or it is thrown at the bin.
- **SOP rule broken:** Step 4, lower the item until it rests flat on the bin floor or on the item below it, then open the gripper.
- **Coaching note:** lay it down, then let go. A dropped item knocks the stack out of line.

Violation: Item carried over a loaded bin or over the freezer mouth

- **Visible cue:** the **right gripper** carries an item over the loaded bag bin, over the other bin, or over items still on the freezer floor, instead of coming to the bin from the front and down from above.
- **SOP rule broken:** Steps 1.1 and 4.2, carry each item level and low, out through the freezer mouth from the front, and come down onto a bin straight from above without passing over the other bin.
- **Coaching note:** far bin first, then the near one, and never fly over what is already stacked.

Violation: Bin not reseated home and square

- **Visible cue:** a bin ends Step 5 short of the freezer back wall, sitting turned, tilted, touching the other bin, or with part of it past the freezer front lip.
- **SOP rule broken:** Step 5, push each bin straight back until it meets the back wall, square, with clear space between the two bins and nothing past the freezer front lip.
- **Coaching note:** straight back until it stops, then check it is square before you let go.

Violation: More than one item moved at once

- **Visible cue:** a gripper carries two items together, two items move at the same time on the freezer floor or on a sort spot, or a bin is moved while an item is still in the other gripper.
- **SOP rule broken:** Steps 1 to 5, one item at a time, and only one arm moves an object at a time.
- **Coaching note:** one item, one place, one motion. Two at a time is how items land in the wrong bin.

Violation: Door moved off the post during the work

- **Visible cue:** the door swings, slides off the door post, or is knocked by an arm while the items, labels, or bins are being worked, or the door post is pushed out of its place.
- **SOP rule broken:** Steps 1 to 5, the door rests against the door post for the whole of the work and the post is never moved.

- **Coaching note:** come in and out of the freezer straight from the front, and nothing touches the door.

Violation: Door swung round with an item still held or an arm still inside

- **Visible cue:** the **left gripper** swings the door while the **right gripper** is still inside the freezer mouth or is still holding an item, or the door catches an arm, a bin, or an item as it comes round.
- **SOP rule broken:** Step 6.1, check both arms are clear of the freezer mouth before the door is swung.
- **Coaching note:** everything out and empty first, then swing. The door does not wait for a hand.

Violation: Door swung by the wrong gripper or slammed

- **Visible cue:** the **right gripper** swings the door round, the **left gripper** grips it somewhere other than its free edge, or the door is swung fast enough to bang against the freezer.
- **SOP rule broken:** Step 6.1, the **left gripper** closes on the door's free edge and swings it round in one smooth move.
- **Coaching note:** left gripper, free edge, smooth swing. A slammed door shifts the bins you just seated.

Violation: Door not closed and sealed

- **Visible cue:** daylight shows along the seal after the **right gripper** comes off, the door springs back open, the latch does not hold, the press is released before 2 seconds, the **right gripper** presses at an angle or on the door's edge, or the press moves the freezer out of square with the back table edge.
- **SOP rule broken:** Step 6.2, press the door's front face flat and straight back at the free edge side until it meets the freezer all round and the latch holds, hold the press for 2 seconds, and do not move the freezer.
- **Coaching note:** flat face, straight back, hold two seconds, then watch it before you move away. A door left ajar is a failed episode.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped, or a check is made and the fault it finds is left uncorrected — an item on the wrong spot, an item number down, a label with a lifted corner, an item in the wrong bin, a stack out of order, a bin short of the back wall, or a door showing daylight.
- **SOP rule broken:** Steps 1, 2, 3, 4, 5, 6.1, and 6.2, run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped, toppled, or knocked over

- **Visible cue:** an item is dropped on the table or the floor, an item already laid down is knocked off its sort spot or out of a bin, a stack is toppled by an arm or by the door, the label sheet is struck and moved, or the freezer is knocked out of square.
- **SOP rule broken:** Steps 1 to 6, nothing is dropped, toppled, or knocked out of its place, and every gripper leaves the freezer straight out to the front or straight up.
- **Coaching note:** check the path and the landing place before the arm moves, and come out the way you went in.

Violation: Wrong arm used

- **Visible cue:** the **right gripper** touches a bin, the door post, or the door's free edge; the **left gripper** picks up or carries an item, lays an item into a bin, or touches the box spot; or an item is handed from one gripper to the other.
- **SOP rule broken:** Steps 1 to 6, the **right gripper** works the items, the labels, and the door press, the **left gripper** works the bins, the label sheet, the bag hold, and the door swing, and nothing changes hands.
- **Coaching note:** right arm on items, left arm on bins and the door. Nothing crosses and nothing is passed.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the door open or showing daylight, an item still on a sort spot or on the freezer floor, a label still on the sheet, a bin short of the back wall or standing proud of the lip, a stack with a number other than 1 on top, the freezer out of square, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 7, confirm the door, the table, the freezer, and the floor, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read a date number, a bin rail label, a label face, or the door seal**, so which item went where or whether the door shut cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Faulty door:** a door whose latch will not hold, whose seal will not close under a correct press, or that swings shut on its own off the door post.

- **Freezer will not stay put:** a freezer that slides or turns on the table under a correct bin pull or a correct door press.
- **Door post will not hold:** a post that slides under the weight of the open door.
- **Faulty bin:** a bin that will not slide on the freezer floor, that tips when drawn by its front rail, that will not sit square against the back wall, or whose front rail breaks under a correct grip.
- **Defective item:** one that leaks, whose seal opens in a correct grip, that will not lie flat on one face, that slips out of a correct grip, or whose date number cannot be read.
- **Faulty label or label sheet:** a label that tears or folds onto itself in a correct peel, a label that will not stick to a clean label patch, a sheet whose two labels do not name bag 2 and bag 3, or a sheet that has fewer or more than two labels.
- **Bin too small:** a bin that will not hold its three items stacked flat with the top item below the bin rim.
- **Freezer floor too small:** a floor that will not take both bins side by side with clear space between them, or that will not let a bin stand at the front lip and at the back wall.
- **A place turns out to sit outside its arm's comfortable reach,** so the freezer floor, a bin at either position, the label sheet, a sort spot, or the door's free edge cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Take one item off the freezer floor
2. Lay one item flat on its sort spot
3. Hold the label sheet flat and peel one label
4. Press one label onto its bag's label patch
5. Draw one bin forward to the freezer front lip
6. Lay one item flat into its bin
7. Build one bin stack in the order 3, 2, 1
8. Push one bin home to the freezer back wall
9. Swing the door round to the freezer
10. Press the door home until it seals
11. Return both arms home and end the episode